The Erdős–Gyárfás Conjecture

Cycles of Power-of-Two Length, the Markström Graph, and Computational Approaches

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Abstract

The Erdős–Gyárfás Conjecture (1995) asserts that every graph with minimum degree at least 3 contains a cycle whose length is a power of 2. Despite its elementary statement, the conjecture remains open. This paper surveys its origin, develops the necessary graph-theoretic background, and summarises what is currently known. We give a full description of the Markström graph, the smallest planar cubic graph with no cycle of length 4 or 8, and verify its properties with an exact cycle-search program, whose implementation and correctness tests we present. We close with a discussion of why the conjecture is interesting from the point of view of formal language theory.

Note on this revision. This version corrects the June 2026 submission. The construction of the Markström graph, the cycle-search algorithm, and several stated results were wrong in the original and have been replaced by statements that are either proved here, verified by the accompanying program, or taken directly from the cited literature.

1 Historical Background and Motivation

1.1 Paul Erdős and the Art of Conjecture

Paul Erdős (1913–1996) was one of the most prolific mathematicians of the twentieth century, with around 1,500 papers across combinatorics, number theory, graph theory and probability. He worked collaboratively, travelled constantly, and often attached small monetary prizes to problems he found important. The *Erdős number*, the collaborative distance from him, has become part of mathematical culture.

1.2 András Gyárfás

András Gyárfás is a Hungarian combinatorialist known for his work in Ramsey theory, graph colouring and the structure of cycles in graphs. He and Erdős formulated the conjecture discussed here in 1995; it appeared in print in Erdős' 1997 problem paper [3], and is listed today as Problem #64 in the Erdős problems database [4].

1.3 Why Powers of Two?

Minimum degree 3 is a weak condition, but it already forces rich cycle structure. It forces cycles, and in fact even cycles (Proposition 3.4). It does not force any *particular* even length: there are cubic graphs with no C_4 (for example the Petersen graph), and with neither C_4 nor C_8 (the Markström graph of Section 4). The powers of two form a sparse, infinite set of lengths, and the conjecture asks whether minimum degree 3 is already enough to hit it.

2 Graph-Theoretic Preliminaries

2.1 Basic Definitions

Definition 2.1 (Graph)**.** A *simple graph* is a pair $G = (V, E)$ where V is a finite set of vertices and $E \subseteq \binom{V}{2}$ is a set of edges. We write $n = |V|$ and $m = |E|$.

Definition 2.2 (Degree)**.** The *degree* $\deg(v)$ of $v \in V$ is the number of edges incident to v. The *minimum degree* of G is $\delta(G) = \min_{v \in V} \deg(v)$.

Definition 2.3 (Cycle)**.** A *cycle* of length $k \geq 3$, written C_k, is a sequence of k *distinct* vertices $v_1, \ldots, v_k$ such that $v_i v_{i+1} \in E$ for all i, indices taken modulo k.

Definition 2.4 (Power-of-two cycle)**.** A cycle is a *power-of-two cycle* if its length lies in $\{4, 8, 16, 32, \dots\} = \{2^k : k \geq 2\}$.

Definition 2.5 (Regular graph)**.** G is *k-regular* if $\deg(v) = k$ for every v; 3-regular graphs are called *cubic*.

2.2 Classical Results on Cycle Lengths

Theorem 2.6 (Dirac, 1952 [2])**.** *If $n \geq 3$ and $\delta(G) \geq n/2$, then G contains a Hamiltonian cycle.*

Theorem 2.7 (Bondy–Simonovits, 1974 [1])**.** *For every $k \geq 2$ there is a constant c_k such that every graph on n vertices with more than $c_k\, n^{1+1/k}$ edges contains a cycle of length exactly $2k$.*

These illustrate two paradigms: large minimum degree forcing long cycles, and edge density forcing a prescribed even length. The Erdős–Gyárfás Conjecture belongs to a third: a *small, constant* minimum degree forcing some length from an infinite, sparse family.

3 The Erdős–Gyárfás Conjecture

3.1 Statement

Conjecture 3.1 (Erdős–Gyárfás, 1995)**.** *Every graph G with $\delta(G) \geq 3$ contains a cycle whose length is a power of 2.*

Example 3.2 (The degree bound is tight)**.** The odd cycle C_{2k+1} has $\delta = 2$ and its only cycle has odd length, so it contains no power-of-two cycle. Hence $\delta \geq 3$ cannot be weakened to $\delta \geq 2$.

Remark 3.3. Finiteness matters: an infinite tree in which every vertex has degree 3 has no cycles at all [4].

3.2 A Warm-Up: Minimum Degree 3 Forces an Even Cycle

The following classical argument shows that $\delta \geq 3$ does force *some* even cycle. It is instructive to see exactly where it stops short of a power of two.

Proposition 3.4. *Every graph with $\delta(G) \geq 3$ contains a cycle of even length.*

Proof. Let $P = x_0 x_1 \cdots x_\ell$ be a longest path in G. Every neighbour of x_0 lies on P, since otherwise P could be extended. As $\deg(x_0) \geq 3$, the vertex x_0 has neighbours x_1, x_i and x_j with $1 < i < j$. These give three cycles:

$$x_0 x_1 \cdots x_i x_0 \text{ (length } i+1), \qquad x_0 x_1 \cdots x_j x_0 \text{ (length } j+1), \qquad x_0 x_i \cdots x_j x_0 \text{ (length } j-i+2).$$

If the first two both have odd length, then i and j are both even, so $j - i + 2$ is even. In every case one of the three cycles is even. □

The proof controls the *parity* of a cycle but says nothing about its exact length, which is precisely what the conjecture needs.

3.3 Current Status

The conjecture is open. The following partial results are known (see [4] for a maintained list).

- **3-connected cubic planar graphs.** Heckman and Krakovski [5] proved the conjecture for this class. The proof is long and partly computer-assisted, using the discharging method.

- **Planar claw-free graphs:** proved by Daniel and Shauger (2001) [4].
- **Large minimum degree.** Liu and Montgomery [6] proved that the conclusion holds whenever the minimum degree is at least some (very large) absolute constant. Erdős and Gyárfás had in fact suspected the opposite: that for every r there is a graph with $\delta \geq r$ and no power-of-two cycle. Liu and Montgomery disproved that stronger suspicion. The gap between their constant and 3 is what remains open.
- **Restricted classes:** the conjecture is known for several families of Cayley graphs, for P_8-free and P_{10}-free graphs, and for graphs of diameter 2 [4].

Size of a counterexample. Exhaustive computer searches by Royle and Markström show that a counterexample must have at least 17 vertices, and a *cubic* counterexample at least 30 vertices [7, 4].

4 The Markström Graph

4.1 Definition

While searching for small counterexamples, Markström [7] showed that the smallest cubic graphs with no C_4 and no C_8 have 24 vertices, and that there are exactly four of them. Exactly one of the four is planar; it is called the *Markström graph M* [10].

Construction. Let H_7 be the graph on vertices $\{0, \ldots, 6\}$ with edges

$$0\text{–}1,\ 0\text{–}2,\ 0\text{–}3,\ 1\text{–}2,\ 1\text{–}4,\ 3\text{–}5,\ 3\text{–}6,\ 4\text{–}5,\ 5\text{–}6.$$

It consists of two triangles $\{0, 1, 2\}$ and $\{3, 5, 6\}$ joined through the vertices 0–3 and 1–4–5. Vertices 2, 4 and 6 have degree 2; we call them the *ports*. Now take the complete graph K_4, replace three of its vertices by copies of H_7 and the fourth by a triangle, and attach each of the original edges of K_4 to one port. This construction follows Exoo [8]. We checked by computer every way of assigning the edges of K_4 to the ports: exactly one result, up to isomorphism, is cubic, planar and free of C_4 and C_8. Since it has 24 vertices, it is one of Markström's four graphs, and since it is planar, it is M. Its cycle spectrum also matches the published one [10] (Theorem 4.3).

The full edge list used in this paper, with vertices $0, \ldots, 23$, is:

H_7 copy A (0–6)	0–1, 0–2, 0–3, 1–2, 1–4, 3–5, 3–6, 4–5, 5–6
H_7 copy B (7–13)	7–8, 7–9, 7–10, 8–9, 8–11, 10–12, 10–13, 11–12, 12–13
H_7 copy C (14–20)	14–15, 14–16, 14–17, 15–16, 15–18, 17–19, 17–20, 18–19, 19–20
Triangle (21–23)	21–22, 21–23, 22–23
Connecting edges	2–11, 4–16, 6–21, 9–18, 13–22, 20–23

Proposition 4.1. *M is cubic, with $|V| = 24$ and $|E| = 36$.*

Proof. Each copy of H_7 has four vertices of degree 3 and three ports of degree 2; each port receives exactly one connecting edge. Each triangle vertex has degree 2 inside the triangle and receives one connecting edge. So every vertex has degree 3, and by the handshaking lemma $|E| = \frac{1}{2} \cdot 24 \cdot 3 = 36$. □

Proposition 4.2. *M is planar, with 14 faces: seven triangles, three pentagons and four faces of length 9.*

Proof. Figure 1 is a drawing without crossings. Counting its faces gives $7 + 3 + 4 = 14$, and indeed $n - m + f = 24 - 36 + 14 = 2$, as Euler's formula requires. □

4.2 Key Property: No C_4 and No C_8

Theorem 4.3 (Markström, 2004; verified in Section 5)**.** *The Markström graph contains no cycle of length* 4 *or* 8*, but it does contain a cycle of length* 16*. Its full cycle spectrum is*

$$\{3, 5, 6, 7\} \cup \{9, 10, \ldots, 24\}.$$

So M avoids the two smallest powers of two, but it is *not* a counterexample: the cycle

$$0 \to 1 \to 2 \to 11 \to 8 \to 7 \to 9 \to 18 \to 15 \to 14 \to 17 \to 20 \to 23 \to 21 \to 6 \to 3 \to 0$$

has length 16. One can check by hand, using the edge list above, that each consecutive pair is an edge and that no vertex repeats.

Remark 4.4. Note that M contains seven triangles. So the presence of a triangle does *not* force a C_4, even in a cubic graph. M is also 3-connected, and so it falls under the theorem of Heckman and Krakovski. It is consistent with that theorem: it has a C_{16}.

4.3 Visualisation

Figure 1 is a Tutte (barycentric) drawing [9]: one face of length 9 is fixed as a regular polygon, and every other vertex sits at the average of its neighbours. For a 3-connected planar graph, this always gives a drawing without crossings, with convex faces.

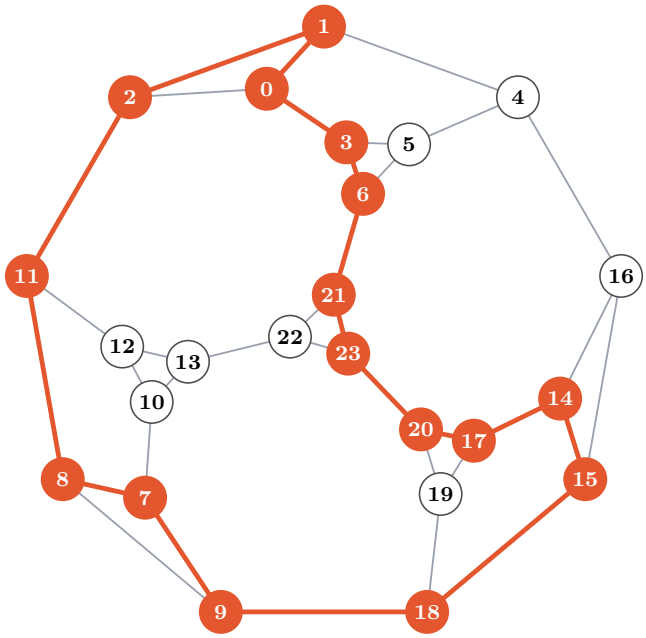


Figure 1: The Markström graph ($n = 24$, $m = 36$, cubic, planar). The highlighted vertices and edges form the 16-cycle from Theorem 4.3. The three copies of H_7 sit at the top, bottom left and bottom right, and the central triangle $\{21, 22, 23\}$ connects them.

5 Computational Approach and Python Implementation

5.1 Overview

The accompanying program (repository linked on the title page) does three things:

1. builds the Markström graph and other test graphs from explicit edge lists, or reads any graph from a text file;

2. for each power of two $k \le n$, decides *exactly* whether G has a k-cycle, and if so prints one as a witness;

3. reports whether the conjecture holds for G.

5.2 The Search Algorithm

To find a cycle of length k, the program runs a depth-limited backtracking search from every start vertex s. It only steps to vertices larger than s, so that each cycle is found from exactly one start, namely its smallest vertex. It keeps track of the vertices on the *current path* and un-marks them when it backtracks.

The second point is essential. The original version of this program used a single "visited" set per start vertex. That explores only one DFS tree, so it misses most cycles. On the Petersen graph, for example, it fails to find the 8-cycle, and on K_4 it fails to find the 4-cycle.

Listing 1: Exact cycle search (`src/erdos_gyarfas/cycles.py`, excerpt).

```
def find_cycle_of_length(graph, k):
    """Return one simple cycle with exactly k vertices, or None."""
    if k < 3 or k > graph.n:
        return None
    adj = [sorted(nbrs) for nbrs in graph.adj]

    for start in range(graph.n):
        if len(adj[start]) < 2:
            continue
        path = [start]
        on_path = [False] * graph.n
        on_path[start] = True
        iters = [iter(adj[start])] # explicit DFS stack
        while iters:
            advanced = False
            for w in iters[-1]:
                if w < start:
                    continue # start is the cycle's minimum
                if w == start:
                    if len(path) == k:
                        return list(path)
                    continue
                if on_path[w] or len(path) == k:
                    continue
                path.append(w) # go deeper
                on_path[w] = True
                iters.append(iter(adj[w]))
                advanced = True
                break
            if not advanced: # backtrack
                iters.pop()
                on_path[path.pop()] = False
    return None
```

5.3 Sample Output

Running `python -m erdos_gyarfas` produces the following (abridged):

```
markstrom
vertices: 24 edges: 36 min degree: 3
C4 none
C8 none
C16 found 0 -> 1 -> 2 -> 11 -> 8 -> 7 -> 9 -> 18 -> 15 -> 14
            -> 17 -> 20 -> 23 -> 21 -> 6 -> 3 -> 0
```

```
-> conjecture holds for this graph

petersen
vertices: 10 edges: 15 min degree: 3
C4 none
C8 found 0 -> 1 -> 2 -> 3 -> 4 -> 9 -> 7 -> 5 -> 0
-> conjecture holds for this graph

k4
vertices: 4 edges: 6 min degree: 3
C4 found 0 -> 1 -> 2 -> 3 -> 0
-> conjecture holds for this graph

c7
vertices: 7 edges: 7 min degree: 2
C4 none
-> conjecture does not apply (min degree < 3)
```

5.4 Correctness Checks

The program comes with an automated test suite. It checks that:

- M is cubic, planar, has 24 vertices and 36 edges, no C_4 or C_8, and the exact cycle spectrum stated in Theorem 4.3, which matches the published one [10];
- K_4, K_5, the Petersen graph, the Heawood graph, the 3-cube and C_7 have their known cycle spectra (for instance, the Petersen graph has cycles of lengths exactly $5, 6, 8, 9$);
- on randomly generated small graphs, the result agrees with a brute force check of every vertex subset and every ordering of it.

In addition, a separate script re-derives M from the K_4/H_7 description alone. It enumerates every gadget matching the description and every way of wiring it, and finds exactly one cubic planar graph with no C_4 and no C_8.

5.5 Complexity

Deciding whether a graph has a cycle of length exactly k is NP-complete in general: the case $k = n$ is the Hamiltonian cycle problem. The search above therefore takes exponential time in the worst case, roughly $O(n \cdot \Delta^{k-1})$ for maximum degree Δ. For the graphs in this project, which have at most a few dozen vertices, every query finishes in milliseconds. Searching for a counterexample at the frontier (30 or more vertices, over all cubic graphs) needs far more sophisticated tools than this program.

6 Connections to Formal Language Theory

From the perspective of the LFA course, the conjecture has a natural reading in terms of languages over a one-letter alphabet.

- **Lengths as unary words.** Identify the number k with the word a^k. The cycle spectrum of G becomes a language $L_G \subseteq a^*$, and the conjecture says that $L_G \cap L_2 \neq \emptyset$ whenever $\delta(G) \geq 3$, where
$$L_2 = \{a^{2^k} : k \geq 2\}.$$
- L_2 **is not regular.** Every regular language over a one-letter alphabet has an ultimately periodic set of lengths. The gaps between consecutive powers of two grow without bound,

so L_2 is not ultimately periodic. Equivalently, the pumping lemma fails: pumping a^{2^k} with $2^k \geq p$ by a block of length $1 \leq j \leq p$ gives a^{2^k+j}, and $2^k < 2^k + j < 2^{k+1}$. So no finite automaton recognises the target set. This is one way to see why arguments that only track a bounded amount of information, such as the parity argument of Proposition 3.4, cannot settle the conjecture on their own.

- **Graphs as automata.** A graph can be read as a nondeterministic automaton over $\{a\}$, with vertices as states and edges as transitions. Its closed walks from a vertex v to itself form a regular language. Cycles, however, must not repeat vertices, and the set of *simple* cycle lengths is not described by the automaton's language. This gap between walks and cycles is exactly what makes cycle problems hard.
- **Search and the stack.** The search procedure in Section 5 keeps its current path on an explicit stack, pushing when it goes deeper and popping when it backtracks, much like a pushdown automaton. Unlike a PDA, though, it also needs random access to the path to test whether a vertex has already been used.

7 Conclusion

The Erdős–Gyárfás Conjecture is easy to state yet sits at the frontier of graph theory. In this paper we:

1. placed the conjecture in its historical context;
2. developed the necessary background, and proved that minimum degree 3 forces an even cycle, though not a cycle of any particular length;
3. summarised the known partial results, including the planar cubic case and the large-minimum-degree theorem of Liu and Montgomery;
4. gave an explicit construction of the Markström graph, drew it, and verified its key properties;
5. presented and tested an exact search program for power-of-two cycles;
6. related the problem to unary languages and the non-regularity of $\{a^{2^k}\}$.

The case of small minimum degree, and of cubic graphs in particular, remains wide open. Its resolution will likely need new structural ideas, or a computer-assisted search far larger than any carried out so far.

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